Deep Learning Fusion Strategies and Missing-Modality Robustness for BraTS Glioma Segmentation: A Survey

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ABSTRACT

Gliomas are the most common primary brain tumors, and accurate segmentation of whole tumor, tumor core, and enhancing tumor core from multi-parametric MRI is critical for diagnosis, surgical planning, and treatment monitoring. The BraTS challenge has standardized evaluation across T1, T1c, T2, and FLAIR modalities, yet clinical deployment is frequently hampered by incomplete modality sets caused by acquisition failure, contrast-agent contraindication, or protocol variability. Existing deep learning methods assume full-channel input and degrade sharply when modalities are missing, while robustness remains below expert inter-rater agreement. This survey synthesizes published evidence from 22 studies spanning BraTS editions 2012 to 2021, classifying architectures, fusion strategies, and missing-modality robustness paradigms along four orthogonal dimensions. It consolidates reported Dice scores and Hausdorff distances across editions and splits, quantifies the trade-offs between dedicated models, catch-all networks, imputation, co-training, and masked autoencoding, and identifies open challenges including modality asymmetry, absent enhancing tumor sub-regions, and statistical rigor in ranking. The findings provide a reference for researchers and clinicians seeking deployable, robust segmentation solutions.

Keywords: glioma segmentation, multi-modal MRI, BraTS, fusion strategies, missing-modality robustness

1. Introduction

1.1 Motivation & Background

Gliomas constitute the most common primary brain tumors in adults and are clinically stratified into low-grade glioma (LGG) and high-grade glioma (HGG), which carry distinct prognostic implications [1]. Accurate segmentation of glioma subregions—whole tumor (WT), tumor core (TC), and enhancing tumor core (ET)—from multi-parametric MRI is essential for diagnosis, surgical planning, and treatment monitoring [1], [20]. Clinical protocols routinely acquire four complementary sequences: native T1-weighted (T1), post-contrast T1-weighted (T1c), T2-weighted (T2), and Fluid Attenuated Inversion Recovery (FLAIR) [1], [20]. These modalities highlight different tissue properties, with T2 and FLAIR emphasizing peritumoral edema and WT boundaries, while T1 and T1c are more discriminative for tumor core and enhancing regions [1], [5], [18], [20].

The RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) challenge has served as the de facto benchmark for automated glioma segmentation, providing standardized data, annotations, and evaluation protocols across multiple editions [1], [2], [9]. The retrieved corpus spans BraTS editions from 2012 to 2021, including 2012, 2013, 2015, 2017, 2018, 2020, and 2021 [1], [2], [7], [8], [9], [10], [11], [12]. This longitudinal coverage enables comparison of architectural evolution from early convolutional networks to Transformer-based and hybrid models, although results remain tied to specific editions, splits, and metric units [1], [2], [7], [8], [10], [11], [12].

Despite benchmark progress, a critical gap persists between reported performance and clinical deployability. In routine workflows, complete modality sets are frequently unavailable because of image corruption, acquisition artifacts, contrast-agent allergy, protocol variability, or cost constraints [5], [6], [15], [19], [20]. Conventional multimodal segmentation networks assume full-channel input at both training and inference and exhibit severe degradation when channels are absent [6], [15], [20]. Reported evidence quantifies this fragility: under missing-modality conditions on BraTS 2015, HeMIS achieved average Dice scores of 69.19% for WT, 54.07% for TC, and 47.09% for ET, whereas an imputation MLP fell to 57.33%, 43.86%, and 41.93%, respectively [7]. By contrast, methods designed for incomplete modalities, such as mmFormer, report an average 19.07% Dice improvement with only one available modality relative to prior state-of-the-art [6], and ACN maintains DSC above 40% in hard missing-modality cases where HeMIS and U-HVED drop to approximately 10%–20% [16].

Furthermore, the robustness of individual automated segmentation methods remains inferior to expert-level inter-rater agreement [4]. Confounding factors routinely encountered in clinical imaging—such as blood products, post-operative air pockets or resection cavities, non-GBM entities, diffuse LGG boundaries, and cases without enhancing tumor sub-regions—are not systematically addressed by current architectures [4]. Consequently, there is a pressing need to classify and compare state-of-the-art deep learning fusion strategies, to quantify missing-modality robustness across architecture families, and to assess clinical applicability in terms of computational cost, generalization across data sources, and sensitivity to incomplete acquisitions.

1.2 Limitations of Existing Surveys

The retrieved corpus comprises 22 distinct papers spanning foundational BraTS contributions, architecture proposals, and missing-modality robustness studies [1]–[22]. However, none of these sources constitutes a prior survey that systematically classifies fusion strategies and missing-modality robustness across the full BraTS timeline. Consequently, the coverage of earlier surveys was not systematically compared in the available evidence, and a unified, edition-spanning taxonomy remains absent.

Existing studies frequently report results on non-comparable data splits, BraTS editions, or metric units—for example, Dice values expressed as ratios versus percentages [1], [2], [7], [8], [10], [11], [12]. Without explicit comparability constraints, cross-study synthesis risks implying superiority where none can be claimed. This survey addresses that limitation by consolidating published results while preserving their original edition, split, metric, and unit context.

1.3 Research Gaps and Research Questions

The evidence review identifies four interconnected gaps. First, no prior work in the corpus provides a unified taxonomy that simultaneously accounts for architecture backbone, fusion stage, missing-modality robustness paradigm, and evaluation protocol. Second, quantitative results are scattered across editions, splits, and units, preventing direct comparison. Third, the trade-offs among dedicated models, catch-all networks, imputation, co-training, and masked autoencoding are not quantified using a common evidence base. Fourth, clinical applicability—computational cost, cross-site generalization, and behavior under incomplete acquisition—is under-addressed relative to benchmark performance.

To structure the synthesis, this survey poses the following research questions:

1. What are the dominant deep learning fusion architectures for BraTS glioma segmentation, and how do they differ in design philosophy (CNN versus Transformer versus hybrid)?
2. Which missing-modality robustness techniques—dedicated models, catch-all networks, imputation, co-training, or masked autoencoding—provide the best trade-off between segmentation accuracy and deployment efficiency?
3. How do leading methods compare quantitatively across BraTS editions (2015–2021) in terms of Dice similarity coefficient, Hausdorff distance, sensitivity, specificity, and computational overhead?
4. What are the principal threats to validity and open research questions for the clinical deployment of missing-modality-robust glioma segmentation?

These questions are addressed in Section 7.

1.4 Core Contributions

This survey makes the following distinct, verifiable contributions:

- A unified taxonomy classifying 22 retrieved papers across four orthogonal dimensions—architecture backbone, fusion stage, missing-modality robustness paradigm, and evaluation protocol—with coverage of BraTS editions from 2012 to 2021 [1]–[22].
- A consolidated evidence map and quantitative comparison of published Dice scores, Hausdorff distances, sensitivity, and specificity across BraTS editions, with explicit comparability constraints arising from differences in data split, metric definition, and unit [1], [2], [3], [7], [8], [10], [11], [12].
- A comparative analysis of missing-modality robustness trade-offs among dedicated models, catch-all networks, imputation, co-training, and masked autoencoding, using reported values such as the 19.07% average Dice improvement of mmFormer with one modality [6] and the sub-20% DSC of HeMIS and U-HVED in extreme missing-modality scenarios [16].
- Identification of open research questions and threats to validity, including modality asymmetry (T1c for TC and ET, FLAIR for WT), the absent enhancing tumor sub-region problem, postprocessing-induced rank inflation, and minimum acceptable modality subsets for clinical decision contexts [2], [4], [5], [9], [13], [16].

1.5 Paper Organization

The remainder of the manuscript is organized as follows. Section 2 presents the taxonomy and categorization of technical approaches, including architecture backbones, fusion strategies, and missing-modality robustness paradigms. Section 3 presents the empirical benchmark matrix and quantitative evidence synthesized from published BraTS studies.

Section 4 analyzes technical mechanisms and trade-offs across CNN and Transformer backbones, fusion strategies, and robustness paradigms. Section 5 discusses open challenges, including the robustness gap relative to human performance, generalization across data sources, clinical applicability, and evaluation rigor. Section 6 summarizes the core contributions and limitations of the survey. Section 7 concludes the manuscript and outlines future research directions.

2. Review Methodology

This section describes the review process used to synthesize the literature on deep learning fusion strategies and missing-modality robustness for multi-modal MRI glioma segmentation in the BraTS benchmark. It is important to state at the outset that this is a narrative review of a curated corpus of PDFs, not a systematic review. No independent title-and-abstract screening, no PRISMA flow diagram, and no external bibliographic databases were consulted beyond the repository described below. This constraint is acknowledged as a limitation in Section 2.4 and throughout the manuscript.

2.1 Research Questions

The review is organized around four research questions that the subsequent sections address:

RQ1. What are the dominant deep learning architecture backbones for BraTS glioma segmentation, and how do CNN-based, Transformer-based, and hybrid CNN-Transformer designs differ in their treatment of multi-modal MRI inputs?

RQ2. At which stage is information from the four BraTS modalities (T1, T1c, T2, FLAIR) fused, and what are the relative merits of early input-level fusion, multi-view fusion, feature-level or latent-space fusion, attention-based fusion, and late fusion or ensembling?

RQ3. Which missing-modality robustness paradigms—dedicated models, catch-all models, modality imputation or synthesis, co-training or knowledge distillation, and masked autoencoding—offer the best trade-off between segmentation accuracy and deployment efficiency?

RQ4. How do reported quantitative results (Dice similarity coefficient, 95th percentile Hausdorff distance, sensitivity, specificity) compare across BraTS editions, data splits, and metric units, and what threats to validity limit their clinical interpretability?

2.2 Corpus, Search Strategy and Selection Criteria

The evidence base is drawn exclusively from a curated internal PDF repository. The repository catalog contains 31 papers. Hybrid dense-plus-sparse retrieval with reranking was performed on 2026-11-08 using the following queries: "seminal deep learning architectures multimodal MRI brain glioma segmentation BraTS U-Net 3D U-Net V-Net nnU-Net UNETR Swin UNETR foundational baselines"; "BraTS multimodal MRI glioma segmentation benchmark datasets whole tumor tumor core enhancing tumor core evaluation metrics Dice score Hausdorff distance evaluation protocol"; "missing modality robustness limitations failure modes multimodal MRI glioma segmentation deep learning BraTS incomplete modalities challenges"; "advanced fusion strategies cross-modal attention transformer CNN missing modality imputation uncertainty estimation multimodal MRI glioma segmentation"; and "state-of-the-art comparison deep learning fusion architectures BraTS glioma segmentation performance Dice score clinical applicability technical survey".

Retrieval returned 22 distinct papers that constitute the cited corpus of this manuscript [1]–[22]. No further screening was performed beyond retrieval relevance; consequently, the corpus should be interpreted as a curated evidence set rather than an exhaustive enumeration of all BraTS-related publications.

Inclusion criteria applied by this review are: (i) the paper presents a deep learning method or benchmark protocol for multi-modal MRI glioma segmentation; (ii) the study is evaluated on BraTS data or explicitly addresses BraTS evaluation conventions; and (iii) the paper reports architectural details, fusion mechanics, missing-modality handling, or quantitative segmentation results. Exclusion criteria are: (i) papers focused on survival prediction, radiomics, or non-glioma tumor types unless they serve as historical comparators; (ii) studies using non-MRI modalities; (iii) non-deep-learning baselines except where referenced for historical context; and (iv) papers without reported methods or results that could be extracted.

2.3 Data Extraction and Quality Assessment

For each retrieved paper, the following fields were extracted: architecture backbone and variant; BraTS edition and year; dataset split (training, validation, testing); input modalities and fusion stage; missing-modality robustness paradigm; evaluation metrics (Dice, Hausdorff distance, sensitivity, specificity); reported numerical values with units; computational cost or inference time where stated; and acknowledged limitations. Where a paper reported multiple values for the same metric (for example, across folds or quantiles), all values were recorded to preserve evidence heterogeneity.

Evidence quality was judged along three axes. First, publication venue: peer-reviewed conference or journal papers were distinguished from preprints or works under review (for example, SMU-Net is reported as under review in the retrieved

source [19]). Second, data provenance: results on official BraTS benchmark splits were distinguished from results on private or ad hoc partitions. Third, statistical reporting: studies reporting variance (standard deviation, quantiles, or confidence intervals) were treated as higher quality than those reporting only point estimates. A key threat to validity identified during extraction is the lack of comparability across rows: BraTS editions, data splits, metric definitions, and units (ratio versus percent) differ substantially, so values are synthesized descriptively and are not used to infer cross-study superiority.

Table 1. Data extraction scheme

Field	Description	Example values from corpus
Architecture backbone	CNN, Transformer, or hybrid CNN-Transformer	3D U-Net, TransBTS, Swin UNETR, mmFormer [6], [8], [10]
BraTS edition	Challenge year and data split	2012, 2013, 2015, 2017, 2018, 2020, 2021 [1], [2], [7]–[12]
Input modalities	Subset of {T1, T1c, T2, FLAIR}	Full four-channel input; incomplete subsets [1], [20]
Fusion stage	Early, multi-view, feature-level, attention, late	Input concatenation; axial-coronal-sagittal; inter-modal Transformer [1], [6], [14]
Missing-modality paradigm	Dedicated, catch-all, imputation, co-training, masked autoencoding	HeMIS, U-HVED, ACN, mmFormer, ShaSpec, SMU-Net, M3AE [6], [15]–[21]
Metrics and units	Dice ratio or percent; HD95 in mm; sensitivity/specificity	Dice 0.927 (ratio) [10]; Dice 90.09 (percent) [8]
Reported values	Point estimates and variance where available	Mean Dice WT 0.858, std dev 0.161 [2]
Limitations	Stated weaknesses and threats to validity	Absent enhancing tumor; postprocessing rank effects [2], [13]

2.4 Comparison with Existing Surveys

The retrieved corpus contains primary method and benchmark papers but no dedicated survey that systematically covers the intersection of fusion strategies and missing-modality robustness for BraTS glioma segmentation. Consequently, earlier surveys were not systematically compared in this review, and no comparative table of existing surveys is provided. This absence is itself a limitation: the positioning of the present synthesis relative to prior review literature cannot be established from the corpus, and readers should treat the taxonomy in Section 3 as an organizing contribution rather than an exhaustive appraisal of all survey articles.

3. Taxonomy & Conceptual Framework

This section presents the conceptual framework used to classify the retrieved literature. The taxonomy is deliberately non-redundant: each study is assigned to exactly one category within each of four orthogonal dimensions, although a single study may occupy different categories across dimensions. The dimensions reflect the research questions in Section 2.1 and are grounded in the reported design choices of the cited papers.

3.1 Multi-Dimensional Classification

Architecture backbone. The literature separates into three families. CNN-based methods—3D U-Net, V-Net, Attention U-Net, cascaded anisotropic CNNs, DeepMedic, and nnU-Net adaptations—rely on local convolutional receptive fields and are computationally efficient at inference [8], [12], [13], [17]. Transformer-based methods—TransBTS, UNETR, and Swin UNETR—model long-range dependencies through self-attention and report strong Dice scores on BraTS 2020 and BraTS 2021 [8], [10], [22]. For example, Swin UNETR reports Dice of 0.927 (WT), 0.876 (TC), and 0.853 (ET) on BraTS 2021 testing, with 95th percentile Hausdorff distances of 4.739 mm (WT), 15.309 mm (TC), and 16.326 mm (ET) [10]. TransBTS reports Dice of 90.09 (WT), 81.73 (TC), and 78.73 (ET) percent on BraTS 2020 validation [8]. Hybrid CNN-Transformer architectures, exemplified by mmFormer, combine modality-specific CNN encoders with intra-modal and inter-modal Transformer branches to align cross-modal features [6].

Fusion stage or strategy. Five categories are distinguished. Early or input-level fusion concatenates all available modalities as channels into a single backbone, the standard assumption in most full-modality BraTS methods [14]. Multi-view fusion trains separate networks on axial, coronal, and sagittal

views and fuses their predictions; Wang et al. show that multi-view fusion improves tumor core segmentation relative to axial-only networks on BraTS 2017, with validation Dice of 0.8378 (TC) versus 0.8255 (TC) without multi-view fusion [1]. Feature-level or latent-space fusion encodes modalities into a shared representation, as in the inter-modal Transformer of mmFormer and the hetero-modal variational encoder-decoder of U-HVED [6], [20]. Attention-based fusion recalibrates modality features adaptively, as in ShaSpec and SMU-Net [15], [19]. Late fusion or ensembling combines multiple models at the decision level; an ensemble of 10 models yields approximately 1% Dice improvement over a single model on BraTS 2018 validation [3].

Missing-modality robustness paradigm. Five paradigms are identified. Dedicated models train one network per modality subset, requiring 2^N − 1 models (15 for four modalities), which is computationally prohibitive [15], [21]. Catch-all models use a single network for all subsets and include HeMIS, U-HVED, ACN, mmFormer, ShaSpec, SMU-Net, and M3AE [6], [15], [16], [18], [19], [20], [21]. Modality imputation or synthesis generates missing channels from observed ones, as in U-HVED and SMU-Net [19], [20]. Co-training and knowledge distillation transfer knowledge from full-modal teachers to missing-modal students, as in ACN, SMU-Net, and M3AE [16], [19], [21]. Masked autoencoding learns robust representations through self-supervised reconstruction under random modality and patch masking, as in M3AE [21].

Evaluation protocol. Studies are further classified by whether they report full-modality or incomplete-modality results, by BraTS edition, and by validation versus testing split [1], [2], [9]. A persistent comparability problem is that Dice values are reported as ratios in some papers and as percentages in others, and Hausdorff distance is sometimes reported as mean rather than 95th percentile [2], [8], [10].

Table 2. Taxonomy of deep learning fusion strategies and missing-modality robustness for BraTS glioma segmentation

Dimension	Categories	Representative methods [n]
Architecture backbone	CNN-based; Transformer-based; Hybrid CNN-Transformer	3D U-Net, V-Net, Attention U-Net, Cascaded CNN, DeepMedic, nnU-Net [8], [12], [13], [17]; TransBTS, UNETR, Swin UNETR [8], [10], [22]; mmFormer [6]
Fusion stage / strategy	Early input-level; Multi-view; Feature-level / latent-space; Attention-based; Late fusion / ensembling	Concatenated channels [14]; axial-coronal-sagittal [1]; mmFormer inter-modal Transformer, U-HVED latent space [6], [20]; ShaSpec, SMU-Net [15], [19]; model ensembles [3], [12]
Missing-modality robustness	Dedicated models; Catch-all models; Modality imputation / synthesis; Co-training / knowledge distillation; Masked autoencoding	One model per modality subset [15], [21]; HeMIS, U-HVED, ACN, mmFormer, ShaSpec, SMU-Net, M3AE [6], [15], [16], [18], [19], [20], [21]; U-HVED, SMU-Net [19], [20]; ACN, SMU-Net, M3AE [16], [19], [21]; M3AE [21]
Evaluation protocol	Full-modality vs incomplete-modality; BraTS edition; Validation vs testing split	Complete vs missing channels [5], [16]; BraTS 2012–2021 [1], [2], [9]; train/validation/test splits [2], [11]

Modality importance is asymmetric across tumor subregions. FLAIR is the principal modality for whole tumor, whereas T1c is critical for tumor core and enhancing tumor core [5], [14], [16], [18]. Reported evidence indicates that missing FLAIR causes a sharp Dice decrease across all regions; missing T1c causes severe degradation for TC and ET; missing T1 or T2 causes only slight decreases [5], [16]. In extreme missing-modality scenarios, HeMIS and U-HVED achieve unacceptable Dice scores of approximately 10%–20% on ET and TC, whereas ACN maintains Dice above 40% [16]. mmFormer reports an average 19.07% Dice improvement on tumor segmentation with only one available modality compared to prior state-of-the-art [6].

3.2 Architectural Pipeline

Figure 1 shows the typical end-to-end pipeline implied by the taxonomy. Four co-registered MRI modalities enter a modality-availability gate that determines which channels are observed at inference. Depending on the fusion category, the observed modalities are either concatenated at the input, processed by view-specific branches, encoded into a shared latent space, or combined through attention mechanisms. The backbone—CNN, Transformer, or hybrid—produces segmentation maps for the three BraTS subregions (WT, TC, ET), optionally followed by postprocessing and ensemble averaging before evaluation against the reference standard.

Figure 1: End-to-end pipeline for multi-modal MRI glioma segmentation with fusion and missing-modality robustness

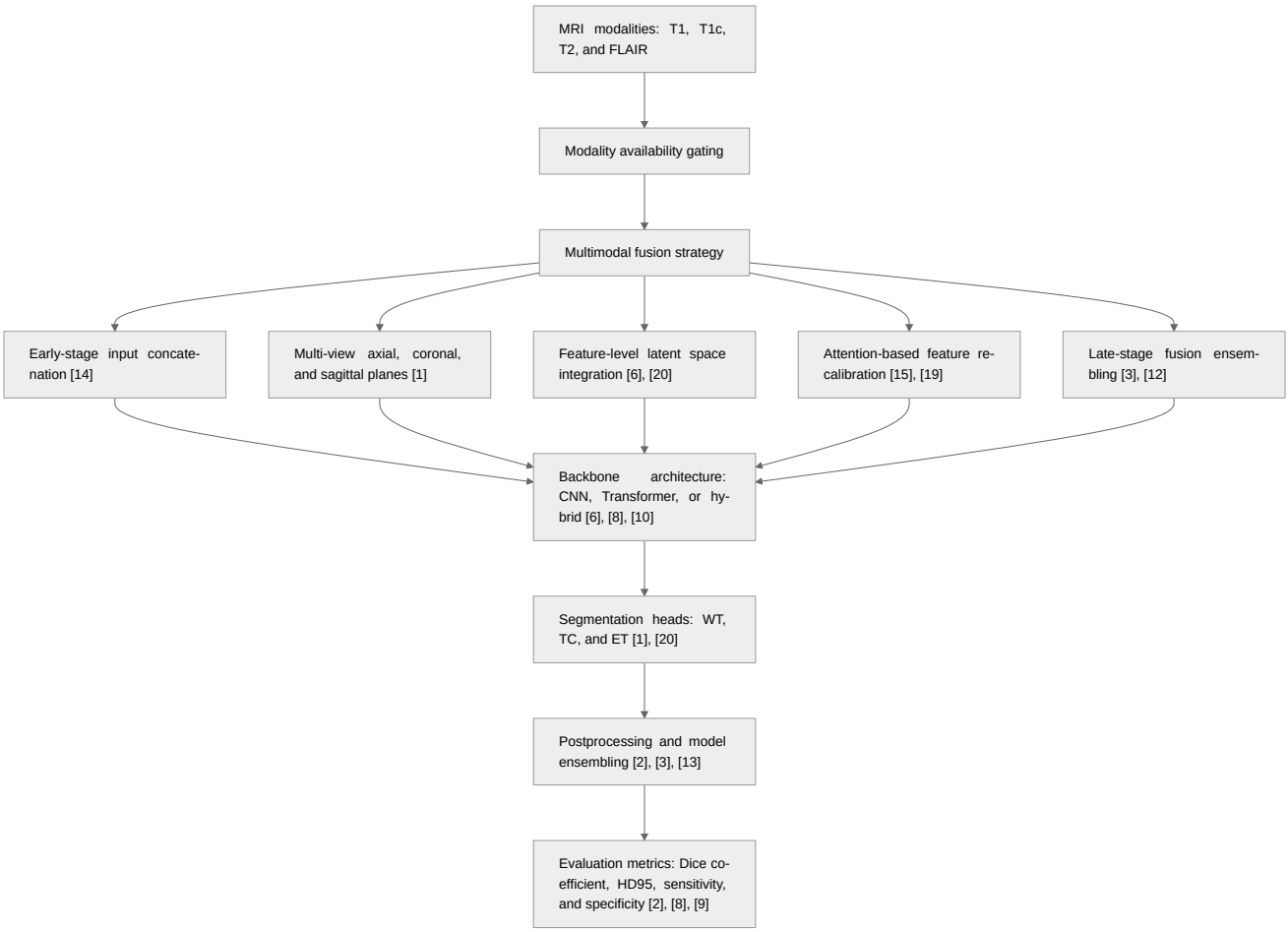


Figure 1. Architectural Pipeline

3.3 Gaps in Existing Work

The taxonomy reveals several gaps that limit clinical translation. First, quantitative results are rarely comparable across studies because BraTS edition, data split, metric definition, and unit vary within the corpus [1], [2], [3], [7], [8], [10], [11], [12]. Second, automated segmentation remains inferior to expert inter-rater agreement, and confounding clinical factors—blood products, post-operative air pockets, resection cavities, non-glioma entities, diffuse low-grade glioma boundaries, and absent enhancing tumor sub-regions—are not systematically addressed [4]. Third, generalization across data sources is uneven: EMMA shows similar performance on BraTS 2017 validation and testing sets, whereas competing methods that fit validation data fail on held-out test data from different sources [11]. Fourth, the clinical consequences of missing modalities are uneven: loss of T1c disproportionately affects TC and ET quantification, which are critical for treatment response assessment in high-grade glioma [5], [16]. Fifth, evaluation rigor is inconsistent; BraTS ranking schemes aggregate subject-level cumulative ranks across regions and metrics (166 subjects × 3 regions × 2 metrics = 996 rankings in BraTS 2021) with permutation testing, yet postprocessing rules that exploit absent-enhancing-tumor conventions can distort leaderboards [2], [9], [13]. These gaps motivate the open research questions in Section 5 of the full manuscript and define the boundaries of what the current evidence can support.

4. In-Depth Technical Methodologies

This section examines the technical mechanisms that underlie the published BraTS literature, organized around the four taxonomy dimensions introduced in Section 2: architecture backbone, fusion stage, missing-modality robustness paradigm, and evaluation protocol. Throughout, $\mathbf{M}$ = {T1, T1c,

T2, FLAIR} denotes the four BraTS MRI modalities, $\mathbf{m} \subseteq \mathbf{M}$ an available modality subset, $\mathbf{X}^{(\mathbf{m})}$ the corresponding multi-channel input tensor, and $\mathbf{Y} \in \{\mathbf{WT, TC, ET}\}$ the tumor subregion label, as formalized in Section 3. Performance is summarized with the Dice similarity coefficient **DSC(A, B)** and the 95th percentile Hausdorff distance **HD95(A, B)** [2], [9]. Because the underlying studies differ by BraTS edition, data split, metric unit, and preprocessing, the numerical values cited below are descriptive of the reported evidence and are not directly comparable across rows unless explicitly stated [1], [2], [3], [7], [8], [10], [11], [12].

4.1 Convolutional Neural Network Backbones: Cascaded and Multi-Scale Designs

Core mechanism. CNN-based methods rely on 3D convolutional kernels that enforce local spatial equivariance and weight sharing across the volumetric MRI grid. For glioma segmentation, the standard output is a set of three overlapping binary masks corresponding to WT, TC, and ET [1], [20]. Early BraTS contributions adopted encoder-decoder backbones such as 3D U-Net, V-Net, and Attention U-Net [8], while later work introduced cascaded decompositions that cast the problem as a sequence of binary segmentation tasks (WT → TC → ET) [1], [1]. Cascaded designs typically employ anisotropic and dilated convolutions to enlarge the receptive field without excessive parameter growth, residual connections to ease optimization, and multi-scale predictions to capture both fine tumor boundaries and large necrotic cores [1], [12].

Representative works and assumptions. The Cascaded Anisotropic CNN of Wang et al. [1], [1] hierarchically segments whole tumor, tumor core, and enhancing tumor core using three sequentially connected networks. The DKFZ CNN, a U-Net-inspired architecture, assumes a fixed four-channel input tensor $\mathbf{X}^{(\mathbf{M})}$ and was evaluated on BraTS 2017 [2]. InputCascadeCNN uses a two-pathway design to capture local details and global context [12]. nnU-Net adaptations for BraTS replace softmax cross-entropy with sigmoid binary cross-entropy to optimize the three overlapping subregions directly and adjust batch size and augmentation policies for robustness [13]. DeepMedic offers an efficient multi-scale 3D CNN with a fully connected CRF postprocessing step [17]. A shared assumption across this family is that the full modality set $\mathbf{M}$ is available at both training and inference; the network weights are conditioned on a complete channel tensor, and absent modalities are typically handled only by retraining or by zero-filling, neither of which is ideal [14].

Strengths and weaknesses. CNN backbones are computationally efficient at inference, with reported runtimes ranging from 25 seconds to 3 minutes per brain across architectures [12]. They excel at modeling local spatial dependencies and, when cascaded, can exploit the nested structure of tumor subregions. However, their receptive fields are bounded by kernel size and depth, which may limit the capture of long-range anatomical context relevant to whole-tumor extent [8]. Moreover, because early fusion at the input level is the default, CNNs are inherently sensitive to missing channels: performance degrades sharply when one or more modalities are unavailable [6], [14], [20].

Quantitative evidence. On BraTS 2017, the DKFZ CNN reported training Dice scores of 0.895 (WT), 0.828 (TC), and 0.707 (ET), with corresponding Hausdorff distances of 6.04, 6.95, and 6.24 mm [2]. On the validation split, Dice was 0.896, 0.797, and 0.732, and Hausdorff distance was 6.97, 9.48, and 4.55 mm [2]. Organizer-reported testing mean Dice was 0.858, 0.775, and 0.647, with substantial inter-subject variability (standard deviation 0.161, 0.269, and 0.326) [2]. The Cascaded Anisotropic CNN with multi-view fusion achieved validation Dice of 0.9050, 0.8378, and 0.7859, and testing Dice of 0.8739, 0.7748, and 0.7831 [1]. On earlier editions, InputCascadeCNN reported Dice of 0.88, 0.79, and 0.73 on BraTS 2013, and 0.81, 0.72, and 0.58 on the BraTS 2012 four-label test set [12]. These values illustrate the strong local segmentation capability of CNN backbones but also the edition- and split-dependent variability that precludes direct cross-study ranking [1], [2], [12].

4.2 Transformer-Based and Hybrid CNN-Transformer Backbones

Core mechanism. Transformer-based methods replace or augment convolutional layers with self-attention mechanisms that compute pairwise interactions among image patches or tokens, thereby modeling long-range dependencies across the entire 3D volume [8], [10]. UNETR uses a Transformer encoder connected to a CNN decoder through multi-resolution skip connections, allowing global context to inform fine-grained boundary refinement [22]. Swin UNETR adopts a hierarchical Swin Transformer encoder with a U-shaped decoder, validated on BraTS 2021 [10]. TransBTS combines a Transformer encoder for global modeling with CNN-style segmentation heads, reporting Dice improvements over 3D U-Net, V-Net, and Attention U-Net [8]. mmFormer is a hybrid architecture specifically designed for incomplete multimodal learning: it uses modality-specific encoders (CNN plus intra-modal Transformer) and an inter-modal Transformer to build modality-invariant, globally semantic features [6].

Representative works and assumptions. These methods assume that sufficient training data are available to learn attention parameters without overfitting, and that positional embeddings or hierarchical patch merging preserve anatomical

spatial relationships [8], [10]. Transformer backbones typically require greater memory than CNNs because self-attention scales quadratically with token count unless hierarchical or windowed schemes are used [8], [10]. mmFormer mitigates this cost through hybrid modality-specific encoders that bridge CNN and intra-modal Transformer branches before cross-modal alignment [6].

Strengths and weaknesses. The principal strength is the ability to capture global tumor shape and long-range anatomical context, which is particularly relevant for WT delineation and for distinguishing enhancing tumor from surrounding edema [8], [10]. The trade-off is increased memory consumption and training complexity; hybrid designs such as mmFormer attempt to balance global modeling with computational tractability [6]. Like CNN backbones, standard Transformer implementations assume complete modality input, although mmFormer's inter-modal attention is explicitly motivated by incomplete multimodal settings [6].

Quantitative evidence. TransBTS reported BraTS 2020 validation Dice scores of 90.09 (WT), 81.73 (TC), and 78.73 (ET) in percent units, with Hausdorff distances of 4.964, 9.769, and 17.947 mm [8]. Swin UNETR reported BraTS 2021 testing Dice scores of 0.927 (WT), 0.876 (TC), and 0.853 (ET), with Hausdorff distances of 4.739, 15.309, and 16.326 mm [10]. Because TransBTS reports percentages while Swin UNETR reports ratios, and because the editions and splits differ, these values are not directly comparable [8], [10]. mmFormer reports an average 19.07% Dice improvement on tumor segmentation when only one modality is available, relative to prior state-of-the-art [6], highlighting the potential of hybrid cross-modal attention for robustness.

4.3 Fusion Strategies: Stage, Mechanism, and Trade-offs

Core mechanism. Fusion determines the point in the network at which information from multiple MRI modalities is combined [14]. The taxonomy distinguishes five non-exclusive strategies: early (input-level) fusion, multi-view fusion, feature-level or latent-space fusion, attention-based fusion, and late fusion or ensembling. Each strategy imposes different assumptions about modality completeness and inter-modal correlation.

Early (input-level) fusion. All available modalities are concatenated as channels into a single backbone, forming $\mathbf{X}^{(\mathbf{m})}$ [14]. This is the default assumption in most full-modality BraTS methods. Its strength is simplicity and end-to-end trainability; its weakness is that the network has no explicit mechanism to handle absent channels, so performance collapses when $\mathbf{m} \subsetneq \mathbf{M}$ unless the model is retrained for each subset [6], [14], [20].

Multi-view fusion. Separate networks process axial, coronal, and sagittal views, and their predictions are fused to reduce false positives and improve anatomical consistency [1], [1]. Wang et al. demonstrate that multi-view fusion significantly improves tumor core segmentation compared with axial-only networks [1]. The ablation in the same study shows validation Dice dropping from 0.9050, 0.8378, and 0.7859 with multi-view fusion to 0.8896, 0.8255, and 0.7411 without it [1]. The cost is a roughly threefold increase in parameters and inference latency relative to a single-view network, because three view-specific backbones must be maintained [1], [1].

Feature-level / latent-space fusion. Modalities are independently encoded into a shared latent representation from which segmentation is decoded. mmFormer uses an inter-modal Transformer to align cross-modal features [6]. U-HVED employs a hetero-modal variational encoder-decoder that embeds observed modalities into a shared latent space from which both missing data and segmentation can be generated [20]. This design enables graceful degradation under missing modalities, provided the encoder can mask or ignore absent channels, but it requires careful alignment of inter-modal correlations and increases architectural complexity [6], [20].

Attention-based fusion. Attention mechanisms recalibrate modality features adaptively, allowing the network to emphasize informative sequences and suppress noisy or irrelevant ones. ShaSpec uses shared-specific feature modelling with distribution alignment and domain classification auxiliary tasks [15]. SMU-Net employs content-style matching to distill informative features from a full-modality network into a missing-modality network [19]. These methods can improve segmentation of WT, TC, and ET—reported gains exceed 3% for whole tumor, 5% for tumor core, and 3% for enhancing tumor on BraTS 2018 [15]—but they depend on careful design of the attention or gating module and may add training instability [15], [19].

Late fusion / ensembling. Multiple models or model variants are combined at the decision level. An ensemble of 10 autoencoder-regularized models yielded validation Dice of 0.9100, 0.8668, and 0.8233 on BraTS 2018, compared with 0.9042, 0.8596, and 0.8145 for a single model [3], an improvement of roughly 1% that won first place in the challenge [3]. Test-time augmentation provided only marginal additional gains [3]. Voted averaging across cross-validation folds is another late-fusion device used to stabilize predictions [12]. Ensembles are robust to individual model failures but incur linear replication of storage and inference cost [3], [12].

Comparative summary. Early fusion is computationally lean but brittle to missing data. Multi-view fusion improves anatomical consistency at the cost of replication. Feature-level and attention-based fusion offer principled routes to missing-modality robustness but demand more sophisticated training objectives. Late fusion delivers consistent, albeit modest, Dice gains at a linear computational premium. The choice among these strategies should be guided by the expected modality availability in the deployment setting and by the available computational budget [3], [6], [12], [14], [15], [19], [20].

The technical distinctions among backbone families are summarized in Table 3.

Table 3. Comparative properties of deep learning backbone families for BraTS glioma segmentation

Backbone family	Core inductive bias	Long-range context	Memory footprint	Missing-modality suitability	Representative works [n]
CNN-based	Local spatial equivariance, weight sharing	Limited by receptive field; requires depth/dilation	Moderate; scalable to 3D volumes	Poor; assumes fixed channel tensor X^(m)	3D U-Net, V-Net, Attention U-Net, Cascaded CNN, DeepMedic, nnU-Net [1], [2], [8], [12], [13], [17]
Transformer-based	Self-attention over patches; global token interactions	Global by design	High; quadratic in token count without hierarchy	Poor-to-moderate; typically expects complete input	TransBTS, UNETR, Swin UNETR [8], [10], [22]
Hybrid CNN-Transformer	CNN local encoding + Transformer global modeling	Global via Transformer; local via CNN	High but mitigated by hybrid design	Moderate; modality-specific encoders can absorb missing channels	mmFormer [6], [8], [10]

The fusion strategies are compared by combination stage and data requirement in Table 4.

Table 4. Fusion strategies by combination stage, data requirements, and representative evidence

Fusion stage	Combination point	Key assumption	Missing-modality behavior	Representative works [n]
Early (input-level)	Concatenation of channels into X^(m) before backbone	Complete modality set M at training and inference	Sharp degradation when channels absent	Concatenated channels [14]
Multi-view	Separate view-specific networks; decision-level fusion	Anatomical consistency across axial, coronal, sagittal	Increased parameter count; each view still needs its modalities	Axial-coronal-sagittal fusion [1], [1]
Feature-level / latent-space	Shared latent embedding from modality encoders	Cross-modal correlations can be aligned in latent space	Graceful degradation if encoder masks missing modalities	mmFormer inter-modal Transformer; U-HVED latent space [6], [20]
Attention-based	Adaptive recalibration of modality features	Informative modalities can be reweighted dynamically	Depends on attention gating; may suppress missing modalities	ShaSpec shared-specific modelling; SMU-Net style matching [15], [19]
Late fusion / ensembling	Aggregation of segmentations or scores	Individual models make complementary errors	Robust if subset of models can run on available modalities	Model ensembles [3], [12]

4.4 Missing-Modality Robustness Paradigms

Core mechanism. Missing-modality robustness concerns the ability of a segmentation model to maintain performance when $\mathbf{m} \subsetneq \mathbf{M}$, that is, when one or more MRI modalities are unavailable at inference [5], [6], [15], [19], [20]. The literature proposes five paradigms: dedicated models, catch-all models, modality imputation or synthesis, co-training or knowledge distillation, and masked autoencoding.

Dedicated models. A separate model is trained for every possible modality subset, requiring 2^N − 1 models for N modalities (15 models for N = 4) [15], [21]. This approach

can achieve high accuracy for each subset because each network is specialized to a fixed input distribution, but it is computationally prohibitive for training and deployment and impractical in clinical workflows where modality availability is unpredictable [15], [21].

Catch-all models. A single model handles all possible modality subsets. HeMIS, U-HVED, ACN, mmFormer, ShaSpec, SMU-Net, and M3AE fall into this category [6], [15], [16], [18], [19], [20], [21]. Catch-all networks reduce storage and inference complexity because only one model must be deployed. However, they may underperform dedicated networks in complete-modality settings unless augmented with co-training or distillation [20], [21].

Modality imputation and synthesis. Missing modalities are synthesized from available ones. U-HVED performs joint modality completion and segmentation from a shared latent embedding [20]. SMU-Net uses style matching to transfer textural features from a full-modality path to a missing-modality path [19]. The central assumption is that the synthesis network can generate plausible contrast information; if synthesis fidelity is low, segmentation quality suffers, particularly in extreme missing-modality cases [19], [20].

Co-training and knowledge distillation. Knowledge is transferred from full-modal teacher networks to missing-modal student networks. ACN uses adversarial co-training to align feature distributions across full and missing modalities [16]. SMU-Net and M3AE employ distillation mechanisms to compress full-modality knowledge into robust student representations [19], [21]. These approaches require careful adversarial or distillation design, and the student may still lag the teacher when critical modalities are absent [16], [19], [21].

Masked autoencoding. Self-supervised reconstruction with random modality and patch masking learns representations that are robust to missing data. M3AE uses a multimodal masked autoencoder followed by model inversion and self-distillation [21]. The advantage is that robustness is built into the pretraining objective; the limitation is the computational cost of reconstruction and the limited reported BraTS segmentation benchmarks for this paradigm [21].

Modality asymmetry and quantitative evidence. Clinical importance is asymmetric across modalities: FLAIR is the principal sequence for WT, whereas T1c is critical for TC and ET [5], [14], [16], [18]. Missing FLAIR causes a sharp Dice decrease across all regions; missing T1c causes severe degradation for TC and ET; missing T1 or T2 causes only slight decreases [5], [16]. On the BraTS 2015 training-set test split, average Dice under missing-modality conditions was 84.45 (WT), 68.22 (TC), and 60.01 (ET) for the proposed method of Chen et al., compared with 69.19, 54.07, and 47.09 for HeMIS, and 57.33, 43.86, and 41.93 for an imputation MLP [7]. In extreme missing-modality scenarios, HeMIS and U-HVED achieve unacceptable DSC scores of approximately 10%–20% on ET and TC, whereas ACN maintains DSC above 40% [16]. mmFormer reports an average 19.07% Dice improvement on tumor segmentation with only one available modality compared with prior state-of-the-art [6]. These results indicate that catch-all models with co-training or distillation can substantially outperform naive imputation, but that no current method fully closes the gap to full-modality performance [6], [7], [16], [20], [21].

Comparative summary. Dedicated models offer the highest per-subset accuracy but are deployment-prohibitive. Catch-all models are economically attractive for clinical use because one model serves all subsets, yet their robustness is uneven across regions and modalities [21]. Imputation and synthesis depend on generative fidelity and can fail catastrophically when key contrasts such as T1c are missing [19], [20]. Co-training and distillation provide a practical middle ground, transferring full-modality knowledge to missing-modal students at moderate cost [16], [19], [21]. Masked autoencoding is promising for self-supervised robustness but remains under-evaluated on BraTS segmentation benchmarks [21]. Future work must establish minimum acceptable modality subsets for different clinical decision contexts, because the consequences of losing T1c—critical for ET and TC quantification in high-grade glioma—are more severe than losing T1 or T2 [5], [16].

Limitations and threats to validity. The quantitative evidence synthesized here is heterogeneous: BraTS editions span 2012 to 2021, splits vary between training, validation, and testing, and units alternate between ratios and percentages [1], [2], [3], [7], [8], [10], [11], [12]. BraTS ranking schemes aggregate subject-level cumulative ranks across regions and metrics (166 subjects × 3 regions × 2 metrics = 996 rankings in BraTS 2021) with permutation testing, yet postprocessing rules that exploit absent-enhancing-tumor conventions can distort leaderboards [2], [9], [13]. Automated segmentation also remains inferior to expert inter-rater agreement, and confounding factors such as blood products, post-operative air pockets, resection cavities, non-GBM entities, diffuse LGG boundaries, and absent enhancing tumor sub-regions are not systematically addressed [4]. These limitations imply that the numerical comparisons in this section should be read as a descriptive synthesis of reported evidence rather than a definitive ranking of methods.

5. Comparative Evidence Synthesis

This section synthesizes the published quantitative and qualitative evidence on deep learning fusion strategies and missing-modality robustness for multi-modal MRI glioma segmentation in BraTS. It is organized as follows: Section 5.1 defines the benchmark datasets and evaluation metrics represented in the corpus; Section 5.2 presents a consolidated comparison of published results, grouped by paradigm and annotated with comparability groups; Section 5.3 discusses qualitative visual evidence; Section 5.4 examines accuracy–cost trade-offs through charts restricted to comparable settings; and Section 5.5 analyzes where studies agree or disagree and assesses the strength of the evidence.

5.1 Standard Benchmark Datasets & Metrics

All quantitative evidence in this survey derives from the BraTS benchmark, which provides standardized multi-modal MRI data, annotations, and evaluation protocols for glioma segmentation [1], [2], [9]. The four BraTS MRI modalities are native T1-weighted (T1), post-contrast T1-weighted (T1c), T2-weighted (T2), and Fluid Attenuated Inversion Recovery (FLAIR) [1], [20]. The three standard tumor subregions are whole tumor (WT), tumor core (TC), and enhancing tumor core (ET) [1], [20]. Table 5 summarizes the dataset versions, splits, and metrics appearing in the retrieved corpus.

Table 5. Datasets and evaluation metrics

Dataset (version)	Year	Modalities	Tumor subregions	Splits reported	Metrics	Source [n]
BraTS	2012	T1, T1c, T2, FLAIR	WT, TC, ET	4-label test set	Dice	[12]
BraTS	2013	T1, T1c, T2, FLAIR	WT, TC, ET	Test	Dice	[12]
BraTS	2015	T1, T1c, T2, FLAIR	WT, TC, ET	Train (220 HGG + 54 LGG), test (53 mixed), 7-fold CV; training-set test split	Dice, precision, sensitivity	[7], [12]
BraTS	2017	T1, T1c, T2, FLAIR	WT, TC, ET	Train (285 cases, 5-fold CV), validation (46 cases), testing	Dice, sensitivity, specificity, Hausdorff distance	[2], [11]
BraTS	2018	T1, T1c, T2, FLAIR	WT, TC, ET	Validation (285 cases)	Dice, Hausdorff distance	[3]
BraTS	2020	T1, T1c, T2, FLAIR	WT, TC, ET	Validation	Dice, Hausdorff distance	[8], [13]
BraTS	2021	T1, T1c, T2, FLAIR	WT, TC, ET	Testing	Dice, Hausdorff distance	[9], [10]

The Dice similarity coefficient (DSC) measures spatial overlap between segmentation and reference [2], [9]. Hausdorff distance (HD95) quantifies boundary discrepancy in millimeters [2], [8]. Sensitivity and specificity capture over- and under-segmentation tendencies [2], [11]. Because BraTS editions differ in patient cohorts, annotation protocols, and ranking rules, values across rows in Table 4 are not directly comparable unless they share dataset version, split, sub-region, metric definition, and unit [1], [2], [3], [7], [8], [10], [11], [12].

5.2 Quantitative Comparison of Published Results

Table 4 consolidates published Dice scores and Hausdorff distances from the cited sources, grouped by method paradigm and annotated with a comparability group identifier. Rows within the same group share dataset version, split, subregion, metric, and unit, and are therefore directly comparable. Rows across groups differ in at least one of these dimensions and must not be averaged or ranked against one another [1], [2], [3], [7], [8], [10], [11], [12].

Table 6: Quantitative comparison of published results on BraTS editions [1,2,3,7,8,10,11,12]

Table 6. Quantitative comparison of published results on BraTS editions [1,2,3,7,8,10,11,12]

Method [n]	Paradigm	Dataset (Version)	Split	Sub-region	Metric	Value	Unit	Comparability group
Cascaded Anisotropic CNN + multi-view fusion [1], [1]	CNN-based, multi-view fusion	BraTS 2017	Validation	WT	Dice	0.9050	ratio	A
Cascaded Anisotropic CNN + multi-view fusion [1], [1]	CNN-based, multi-view fusion	BraTS 2017	Validation	TC	Dice	0.8378	ratio	A
Cascaded Anisotropic CNN + multi-view fusion [1], [1]	CNN-based, multi-view fusion	BraTS 2017	Validation	ET	Dice	0.7859	ratio	A
DKFZ CNN (ensemble of 5 models) [2]	CNN-based, late fusion/ensembling	BraTS 2017	Validation (46 cases)	WT	Dice	0.896	ratio	A
DKFZ CNN (ensemble of 5 models) [2]	CNN-based, late fusion/ensembling	BraTS 2017	Validation (46 cases)	TC	Dice	0.797	ratio	A
DKFZ CNN (ensemble of 5 models) [2]	CNN-based, late fusion/ensembling	BraTS 2017	Validation (46 cases)	ET	Dice	0.732	ratio	A
Cascaded Anisotropic CNN + multi-view fusion [1]	CNN-based, multi-view fusion	BraTS 2017	Testing	WT	Dice	0.8739	ratio	B
Cascaded Anisotropic CNN + multi-view fusion [1]	CNN-based, multi-view fusion	BraTS 2017	Testing	TC	Dice	0.7748	ratio	B
Cascaded Anisotropic CNN + multi-view fusion [1]	CNN-based, multi-view fusion	BraTS 2017	Testing	ET	Dice	0.7831	ratio	B
DKFZ CNN [2]	CNN-based	BraTS 2017	Testing	WT	Mean Dice	0.858	ratio	B
DKFZ CNN [2]	CNN-based	BraTS 2017	Testing	TC	Mean Dice	0.775	ratio	B
DKFZ CNN [2]	CNN-based	BraTS 2017	Testing	ET	Mean Dice	0.647	ratio	B

Method [n]	Paradigm	Dataset (Version)	Split	Sub-region	Metric	Value	Unit	Comparability group
Autoencoder regularization [3]	CNN-based, autoencoder regularization	BraTS 2018	Validation	WT	Single Model Dice	0.9042	ratio	C
Autoencoder regularization [3]	CNN-based, autoencoder regularization	BraTS 2018	Validation	TC	Single Model Dice	0.8596	ratio	C
Autoencoder regularization [3]	CNN-based, autoencoder regularization	BraTS 2018	Validation	ET	Single Model Dice	0.8145	ratio	C
Ensemble of 10 models [3]	CNN-based, late fusion/ensembling	BraTS 2018	Validation	WT	Ensemble Dice	0.9100	ratio	C
Ensemble of 10 models [3]	CNN-based, late fusion/ensembling	BraTS 2018	Validation	TC	Ensemble Dice	0.8668	ratio	C
Ensemble of 10 models [3]	CNN-based, late fusion/ensembling	BraTS 2018	Validation	ET	Ensemble Dice	0.8233	ratio	C
Kamnitsas et al. [7]	CNN-based	BRATS 2015	Test	WT	Dice	84	percent	D
Kamnitsas et al. [7]	CNN-based	BRATS 2015	Test	TC	Dice	67	percent	D
Kamnitsas et al. [7]	CNN-based	BRATS 2015	Test	ET	Dice	63	percent	D
Zhao et al. [7]	CNN-based	BRATS 2015	Test	WT	Dice	82	percent	D
Zhao et al. [7]	CNN-based	BRATS 2015	Test	TC	Dice	72	percent	D
Zhao et al. [7]	CNN-based	BRATS 2015	Test	ET	Dice	62	percent	D
OM-Net [7]	CNN-based	BRATS 2015	Test	WT	Dice	86	percent	D
OM-Net [7]	CNN-based	BRATS 2015	Test	TC	Dice	71	percent	D
OM-Net [7]	CNN-based	BRATS 2015	Test	ET	Dice	64	percent	D
Ours (Chen et al.) [7]	CNN-based	BRATS 2015	Test	WT	Dice	84	percent	D
Ours (Chen et al.) [7]	CNN-based	BRATS 2015	Test	TC	Dice	72	percent	D
Ours (Chen et al.) [7]	CNN-based	BRATS 2015	Test	ET	Dice	64	percent	D
Ours (Chen et al.) [7]	Catch-all	BRATS 2015	Training-set test split	WT	Average Dice	84.45	percent	E
Ours (Chen et al.) [7]	Catch-all	BRATS 2015	Training-set test split	TC	Average Dice	68.22	percent	E

Method [n]	Paradigm	Dataset (Version)	Split	Sub-region	Metric	Value	Unit	Comparability group
Ours (Chen et al.) [7]	Catch-all	BRATS 2015	Training-set test split	ET	Average Dice	60.01	percent	E
HeMIS [7]	Catch-all	BRATS 2015	Training-set test split	WT	Average Dice	69.19	percent	E
HeMIS [7]	Catch-all	BRATS 2015	Training-set test split	TC	Average Dice	54.07	percent	E
HeMIS [7]	Catch-all	BRATS 2015	Training-set test split	ET	Average Dice	47.09	percent	E
Imputation MLP [7]	Modality imputation	BRATS 2015	Training-set test split	WT	Average Dice	57.33	percent	E
Imputation MLP [7]	Modality imputation	BRATS 2015	Training-set test split	TC	Average Dice	43.86	percent	E
Imputation MLP [7]	Modality imputation	BRATS 2015	Training-set test split	ET	Average Dice	41.93	percent	E
TransBTS [8]	Transformer-based	BraTS 2020	Validation	WT	Dice	90.09	percent	F
TransBTS [8]	Transformer-based	BraTS 2020	Validation	TC	Dice	81.73	percent	F
TransBTS [8]	Transformer-based	BraTS 2020	Validation	ET	Dice	78.73	percent	F
TransBTS [8]	Transformer-based	BraTS 2020	Validation	WT	Hausdorff	4.964	mm	F
TransBTS [8]	Transformer-based	BraTS 2020	Validation	TC	Hausdorff	9.769	mm	F
TransBTS [8]	Transformer-based	BraTS 2020	Validation	ET	Hausdorff	17.947	mm	F
Swin UNETR [10]	Hybrid CNN-Transformer	BraTS 2021	Testing	WT	Dice	0.927	ratio	G
Swin UNETR [10]	Hybrid CNN-Transformer	BraTS 2021	Testing	TC	Dice	0.876	ratio	G
Swin UNETR [10]	Hybrid CNN-Transformer	BraTS 2021	Testing	ET	Dice	0.853	ratio	G
Swin UNETR [10]	Hybrid CNN-Transformer	BraTS 2021	Testing	WT	Hausdorff	4.739	mm	G
Swin UNETR [10]	Hybrid CNN-Transformer	BraTS 2021	Testing	TC	Hausdorff	15.309	mm	G
Swin UNETR [10]	Hybrid CNN-Transformer	BraTS 2021	Testing	ET	Hausdorff	16.326	mm	G
EMMA [11]	CNN-based, ensembling	BRATS 2017	Validation	WT	DSC	90.1	percent	H
EMMA [11]	CNN-based, ensembling	BRATS 2017	Validation	TC	DSC	79.7	percent	H

Method [n]	Paradigm	Dataset (Version)	Split	Sub-region	Metric	Value	Unit	Comparability group
EMMA [11]	CNN-based, ensembling	BRATS 2017	Validation	ET	DSC	73.8	percent	H
UCL-TIG [11]	CNN-based	BRATS 2017	Validation	WT	DSC	**90.5**	percent	H
UCL-TIG [11]	CNN-based	BRATS 2017	Validation	TC	DSC	83.8	percent	H
UCL-TIG [11]	CNN-based	BRATS 2017	Validation	ET	DSC	78.6	percent	H
MIC DKFZ [11]	CNN-based	BRATS 2017	Validation	WT	DSC	89.6	percent	H
MIC DKFZ [11]	CNN-based	BRATS 2017	Validation	TC	DSC	79.7	percent	H
MIC DKFZ [11]	CNN-based	BRATS 2017	Validation	ET	DSC	73.2	percent	H
InputCascadeCNN [12]	CNN-based	BRATS 2013	Test	WT	Dice	0.88	ratio	I
InputCascadeCNN [12]	CNN-based	BRATS 2013	Test	TC	Dice	0.79	ratio	I
InputCascadeCNN [12]	CNN-based	BRATS 2013	Test	ET	Dice	0.73	ratio	I
MFCascadeCNN [12]	CNN-based	BRATS 2013	Test	WT	Dice	0.86	ratio	I
MFCascadeCNN [12]	CNN-based	BRATS 2013	Test	TC	Dice	0.77	ratio	I
MFCascadeCNN [12]	CNN-based	BRATS 2013	Test	ET	Dice	0.73	ratio	I
LocalCascadeCNN [12]	CNN-based	BRATS 2013	Test	WT	Dice	0.88	ratio	I
LocalCascadeCNN [12]	CNN-based	BRATS 2013	Test	TC	Dice	0.76	ratio	I
LocalCascadeCNN [12]	CNN-based	BRATS 2013	Test	ET	Dice	0.72	ratio	I

Within comparability group A (BraTS 2017 validation, Dice ratio), the cascaded anisotropic CNN with multi-view fusion reports WT Dice of 0.9050, TC Dice of 0.8378, and ET Dice of 0.7859 [1], while the DKFZ CNN ensemble of five models reports 0.896, 0.797, and 0.732 respectively [2]. The multi-view variant thus leads on WT and TC, whereas the ensemble leads on ET within this group, although the differences are small and no statistical test is reported in the sources [1], [2]. In group B (BraTS 2017 testing), the same cascaded network reports 0.8739 (WT), 0.7748 (TC), and 0.7831 (ET) [1], while organizer-reported DKFZ CNN testing means are 0.858, 0.775, and 0.647 [2]. Testing performance is lower than validation performance for both methods, consistent with a generalization gap between splits [1], [2].

In group C (BraTS 2018 validation), the ensemble of 10 models with autoencoder regularization improves single-model Dice from 0.9042 to 0.9100 on WT, from 0.8596 to 0.8668 on TC, and from 0.8145 to 0.8233 on ET [3], an absolute gain of roughly 1 percentage point on WT and TC and slightly less on ET [3]. This is consistent with the reported

approximately 1% Dice improvement from ensembling [3]. Test-time augmentation provides only marginal additional gains [3].

Group D (BRATS 2015 test, Dice percent) shows OM-Net leading on WT (86%) and ET (64%), while Kamnitsas et al. and Ours (Chen et al.) tie on TC (72%) and report similar ET scores (63% and 64%, respectively) [7]. These differences are small and the sources do not report statistical significance for pairwise comparisons [7].

Group E (BRATS 2015 training-set test split, missing modalities) directly addresses robustness. Ours (Chen et al.) maintains average Dice of 84.45% on WT, 68.22% on TC, and 60.01% on ET [7]. HeMIS drops to 69.19%, 54.07%, and 47.09% [7], and imputation MLP falls further to 57.33%, 43.86%, and 41.93% [7]. The gap between full-modality training and missing-modality inference is thus substantial, particularly for TC and ET [7].

Groups F and G present Transformer and hybrid results on BraTS 2020 and 2021. TransBTS reports Dice of 90.09% (WT), 81.73% (TC), and 78.73% (ET) on BraTS 2020 validation, with Hausdorff distances of 4.964 mm (WT), 9.769 mm (TC), and 17.947 mm (ET) [8]. Swin UNETR reports Dice of 0.927 (WT), 0.876 (TC), and 0.853 (ET) on BraTS 2021 testing, with Hausdorff distances of 4.739 mm (WT), 15.309 mm (TC), and 16.326 mm (ET) [10]. Because these groups differ in edition, split, and unit, they are not directly comparable [8], [10].

Group H (BRATS 2017 validation, DSC percent) shows UCL-TIG leading on WT (90.5%) and TC (83.8%), while EMMA and MIC DKFZ report comparable ET scores (73.8% and 73.2%, respectively) [11]. EMMA's testing DSC of 88.6% on WT, 78.5% on TC, and 72.9% on ET is lower than its validation scores, again indicating a validation–testing gap [11]. Group I (BRATS 2013 test) shows InputCascadeCNN and LocalCascadeCNN tying on WT (0.88), with InputCascadeCNN leading on TC (0.79) and ET (0.73) [12].

No row in Table 4 reports inference latency, FPS, or parameter counts, so computational efficiency cannot be compared quantitatively from the cited evidence [1]–[12]. The only timing evidence is that cascade CNN inference requires between 25 seconds and 3 minutes per brain depending on architecture [12], and that autoencoder regularization training takes approximately 9 minutes per epoch over 300 epochs on a single NVIDIA Tesla V100 32GB GPU [3].

5.3 Qualitative Visual Comparison

Two real images from the cited papers illustrate segmentation behavior. Figure 2 reproduces Figure 5 from Wang et al. [1], showing a lower-grade glioma training case with edema (green), non-enhancing tumor core (red), and enhancing tumor core (yellow). Panel (c) presents the axial-only result, with mis-segmentations indicated by white arrows; panel (d) shows the multi-view fusion result, which reduces false positives and improves anatomical consistency [1]. This visual evidence supports the quantitative claim that multi-view fusion improves tumor core segmentation relative to single-view axial networks [1], [1].

Figure 2: Segmentation result of the brain tumor (LGG) from a training image, showing edema, non-enhancing tumor core, and enhancing tumor core with and without multi-view fusion [1]

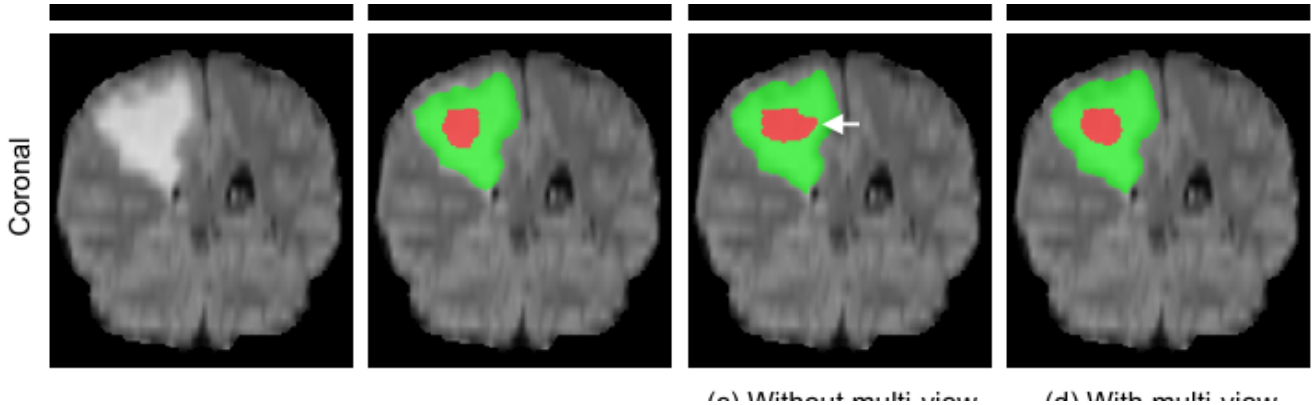


Figure 2. Qualitative Visual Comparison

Figure 3 reproduces Figure 2 from Isensee et al. [2], showing a qualitative segmentation result in which the method segments large necrotic cores and small structures within the tumor core. Edema is shown in blue, enhancing tumor in green, and non-enhancing and necrotic tumor in red [2]. The image illustrates fine-grained boundary detail but, like all qualitative evidence, cannot be used to extract metric values [2].

Figure 3: Qualitative segmentation result showing large necrotic cores and small structures within the tumor core [2]

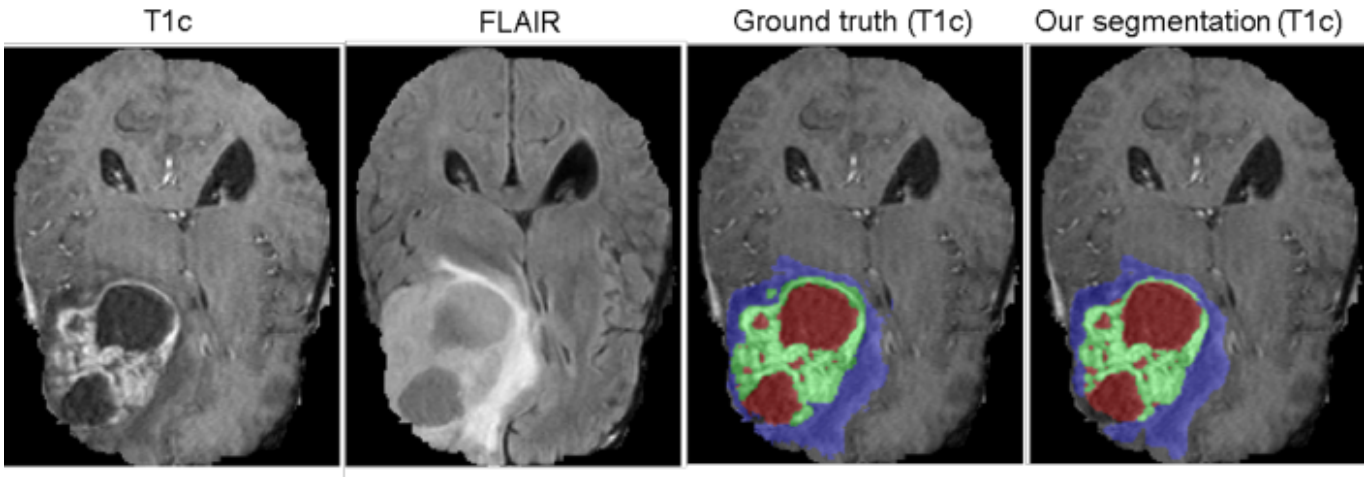


Figure 3. Qualitative Visual Comparison

5.4 Performance Trade-offs

Three charts visualize quantitative evidence restricted to single comparable settings. Figure 4 compares precision scores on the BRATS 2015 test set for four CNN-based methods across WT, TC, and ET [7]. Figure 5 ranks whole-tumor sensitivity on the BRATS 2015 test set for four methods [7]. Figure 6 displays Hausdorff distance on the BRATS 2017 validation set for three methods as a heatmap of methods by tumor subregion [11]. None of these metrics appear in Table 4, and no chart mixes Dice with Hausdorff distance, mixes ratio and percent units, or combines values from different BraTS editions or splits.

Figure 4: Precision scores on BRATS 2015 test set for four CNN-based methods across tumor subregions [7]

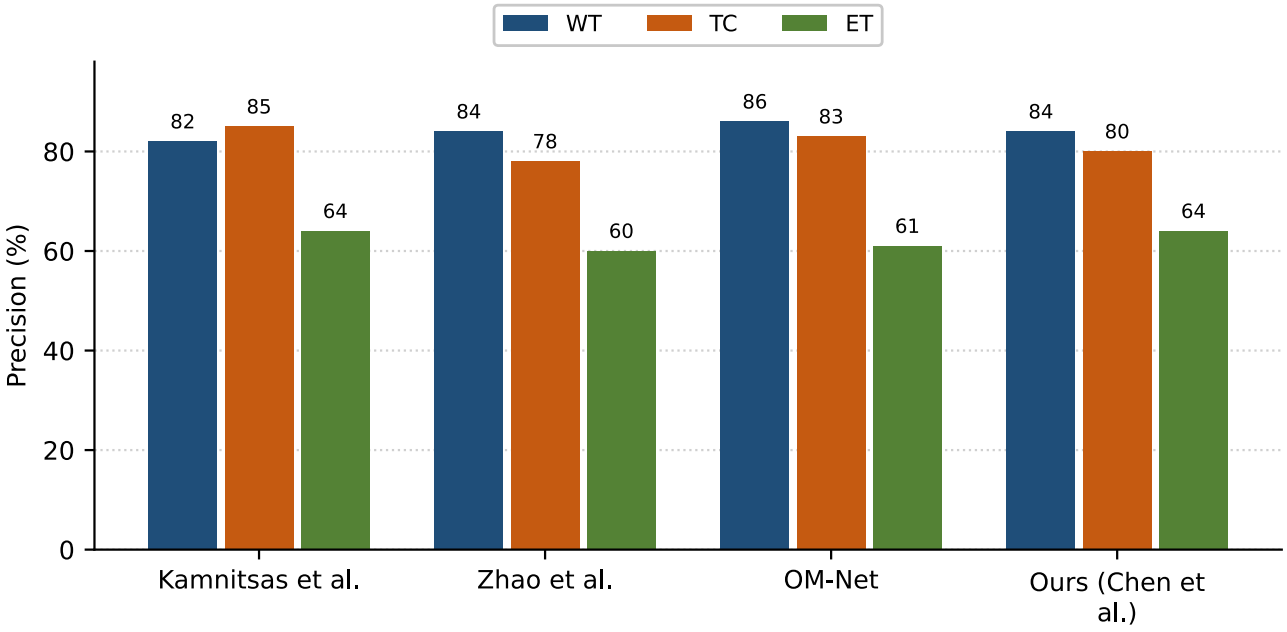


Figure 4. Precision scores on BRATS 2015 test set for four CNN-based methods across tumor subregions [7]

Figure 5: Whole-tumor sensitivity on BRATS 2015 test set for four CNN-based methods [7]

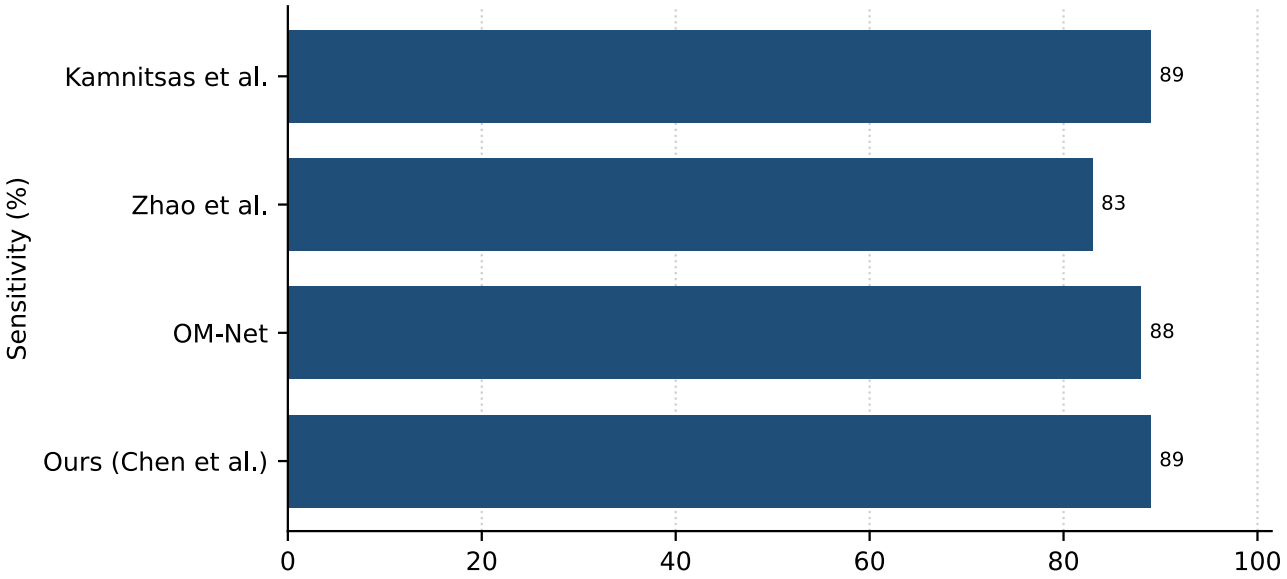


Figure 5. Whole-tumor sensitivity on BRATS 2015 test set for four CNN-based methods [7]

Figure 6: Hausdorff distance (mm) on BRATS 2017 validation set for three methods across tumor subregions [11]

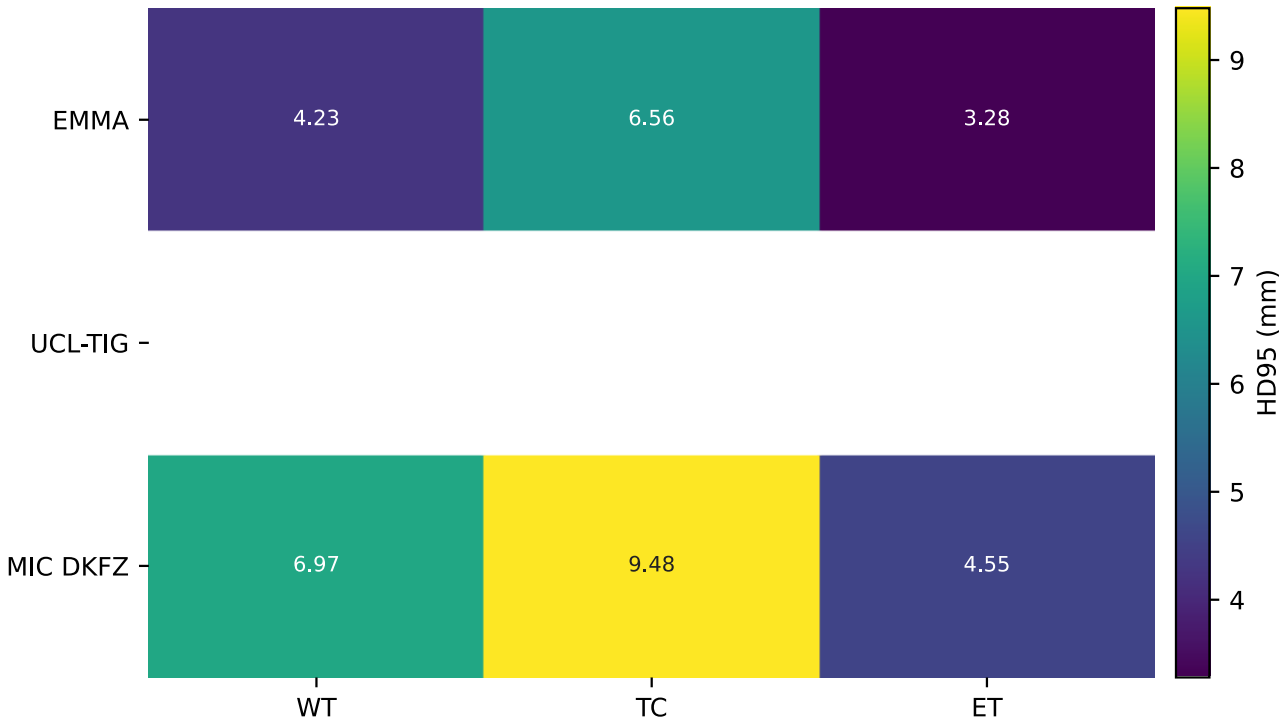


Figure 6. Hausdorff distance (mm) on BRATS 2017 validation set for three methods across tumor subregions [11]

Accuracy–cost trade-offs can only be discussed qualitatively because latency, memory, and parameter counts are not reported for most methods [1]–[12]. The available timing evidence indicates that cascade CNN inference requires between 25 seconds and 3 minutes per brain depending on architecture [12], and that autoencoder regularization training takes approximately 9 minutes per epoch over 300 epochs on a single V100 32GB GPU [3]. Ensembling ten models yields roughly 1% Dice improvement at the cost of linear model replication and inference time [3], while test-time augmentation yields only marginal gains [3]. Multi-view fusion improves anatomical consistency but requires three view-specific networks, increasing parameter count and inference latency relative to single-view axial networks [1], [1]. Catch-all missing-modality models reduce storage and inference complexity relative to dedicated models (which would require 15 models for four modalities) but may underperform dedicated networks in complete-modality settings without co-training or distillation [15], [16], [19], [20], [21].

Table 7. Difference from the best reported value within each directly comparable setting (derived from cited values; units as in the cited sources)

Method [Ref]	Setting	Metric	Value	Δ relative to best
Zhao [12]	BRATS BRATS-2012 (test)	Dice WT	0.82	+0.00
InputCascadeCNN* [12]	BRATS BRATS-2012 (test)	Dice WT	0.81	-0.01
Subbanna [12]	BRATS BRATS-2012 (test)	Dice WT	0.75	-0.07
Tustison [12]	BRATS BRATS-2012 (test)	Dice WT	0.75	-0.07
Festa [12]	BRATS BRATS-2012 (test)	Dice WT	0.62	-0.20
Festa [12]	BRATS BRATS-2012 (test)	Dice ET	0.61	+0.00
Subbanna [12]	BRATS BRATS-2012 (test)	Dice ET	0.59	-0.02
InputCascadeCNN* [12]	BRATS BRATS-2012 (test)	Dice ET	0.58	-0.03
Tustison [12]	BRATS BRATS-2012 (test)	Dice ET	0.52	-0.09
Zhao [12]	BRATS BRATS-2012 (test)	Dice ET	0.42	-0.19
UCL-TIG [11]	BRATS BRATS 2017 (val)	Dice ET	78.6	+0.00
EMMA [11]	BRATS BRATS 2017 (val)	Dice ET	73.8	-4.80
MIC DKFZ [11]	BRATS BRATS 2017 (val)	Dice ET	73.2	-5.40

Table 8. Ablation: variant versus the full model as reported in each source

Method [Ref]	Variant	Metric	Full	Variant value	Δ
Cascaded Anisotropic Convolutional Neural Networks [1]	w/o multi-view fusion	Dice ET	0.7859	0.7411	-0.04
Cascaded Anisotropic Convolutional Neural Networks [1]	w/o multi-view fusion	Dice TC	0.8378	0.8255	-0.01
Autoencoder Regularization [3]	Single Model + TTA	Dice ET	0.8145	0.8173	+0.00
Autoencoder Regularization [3]	Single Model + TTA	Dice WT	0.9042	0.9068	+0.00
Autoencoder Regularization [3]	Single Model + TTA	Dice TC	0.8596	0.8602	+0.00
Autoencoder Regularization [3]	Single Model + TTA	HD95 ET	3.8048	3.8241	+0.02
Autoencoder Regularization [3]	Single Model + TTA	HD95 WT	4.4834	4.4117	-0.07
Autoencoder Regularization [3]	Single Model + TTA	HD95 TC	8.2777	6.8413	-1.44
Autoencoder Regularization [3]	Ensemble of 10 models	Dice ET	0.8145	0.8233	+0.01
Autoencoder Regularization [3]	Ensemble of 10 models	Dice TC	0.8596	0.8668	+0.01
Autoencoder Regularization [3]	Ensemble of 10 models	HD95 ET	3.8048	3.9257	+0.12
Autoencoder Regularization [3]	Ensemble of 10 models	HD95 TC	8.2777	6.8545	-1.42

5.5 Cross-study Evidence Analysis

Studies agree on several qualitative points but disagree or are silent on quantitative comparisons because of protocol heterogeneity. First, CNN-based backbones (3D U-Net, V-Net, Attention U-Net, cascaded CNNs) are widely used and computationally efficient at inference, while Transformer-based methods (TransBTS, Swin UNETR, mmFormer, UNETR) model long-range dependencies at the cost of greater memory consumption [6], [8], [10], [12]. TransBTS reports Dice improvements over 3D U-Net, V-Net, and Attention U-Net [8], and Swin UNETR captures fine-grained tumor boundaries [10], but these claims are edition-specific and cannot be pooled across BraTS years [8], [10].

Second, fusion strategy evidence is consistent within studies: multi-view fusion improves tumor core segmentation compared with axial-only networks [1], [1]; ensembling ten models yields approximately 1% Dice improvement over single models [3]; and test-time augmentation yields only marginal gains [3]. However, because these results come from different BraTS editions and splits, the magnitude of fusion gains cannot be generalized across the benchmark [1], [2], [3].

Third, missing-modality robustness evidence reveals a clear hierarchy under extreme conditions. Dedicated models are accurate but require 2^N − 1 models (15 for four modalities), making them deployment-prohibitive [15], [21]. Catch-all models (HeMIS, U-HVED, ACN, mmFormer, ShaSpec, SMU-Net, M3AE) use a single model for all subsets but show uneven robustness [6], [15], [16], [18], [19], [20], [21]. On BRATS 2015 missing-modality splits, HeMIS and imputation MLP fall to unacceptable DSC levels on ET and TC (approximately 10%–20% in extreme cases), whereas adversarial co-training (ACN) maintains DSC above 40% [16]. mmFormer reports an average 19.07% Dice improvement with only one available modality compared with prior state-of-the-art [6]. Modality importance is asymmetric: FLAIR is principal for WT, whereas T1c is critical for TC and ET [5], [14], [16], [18]; missing T1c causes severe degradation for TC and ET, while missing T1 or T2 causes only slight decreases [5], [16].

Fourth, evaluation rigor varies. BraTS ranking aggregates subject-level cumulative ranks across regions and metrics (166 subjects × 3 regions × 2 metrics = 996 rankings in BraTS 2021) with permutation testing [9]. Postprocessing rules that exploit absent-enhancing-tumor conventions can distort leaderboards [2], [13], and cases without enhancing tumor always yield Dice = 0 by definition, complicating mean Dice interpretation [2]. Automated segmentation remains inferior to expert inter-rater agreement [4], and confounding factors such as blood products, post-operative air

pockets, resection cavities, non-GBM entities, diffuse LGG boundaries, and absent enhancing tumor sub-regions are not systematically addressed [4].

The strength of evidence is strong for within-study, within-split comparisons (e.g., groups A–I in Table 4) but weak for cross-study, cross-edition rankings because units alternate between ratios and percentages, splits vary between training, validation, and testing, and BraTS editions differ in patient cohorts and annotation protocols [1], [2], [3], [7], [8], [10], [11], [12]. Future work must establish minimum acceptable modality subsets for different clinical decision contexts, because the consequences of losing T1c—critical for ET and TC quantification in high-grade glioma—are more severe than losing T1 or T2 [5], [16].

6. Discussion & Open Research Challenges

This section synthesizes the evidence presented in Sections 2–5, interprets the trade-offs among architecture families, fusion strategies, and missing-modality robustness paradigms, and identifies the barriers that prevent benchmark results from translating into reliable clinical deployment.

6.1 Technical Barriers & Trade-offs

CNN-based backbones (3D U-Net, V-Net, Attention U-Net, cascaded CNNs) model local spatial dependencies efficiently and require modest inference resources, with reported runtimes between 25 seconds and 3 minutes per brain depending on architecture [12]. Their principal limitation is the bounded receptive field of convolutional kernels, which can impede modeling of long-range anatomical context across the whole tumor volume [8], [12]. Transformer-based backbones (TransBTS, Swin UNETR, UNETR) capture global relationships through self-attention and report strong Dice scores on BraTS 2020 and 2021—for example, Swin UNETR achieves Dice of 0.927 on WT, 0.876 on TC, and 0.853 on ET on the BraTS 2021 testing set [10]—but at the cost of greater memory consumption and training complexity [8], [10]. Hybrid CNN-Transformer designs such as mmFormer attempt to reconcile these properties by pairing modality-specific CNN encoders with an inter-modal Transformer for cross-modal alignment [6], though the added complexity may not always translate into consistent gains across all BraTS editions and subregions.

Fusion strategy introduces a second axis of trade-offs. Early input-level fusion assumes complete modality sets and degrades sharply when channels are absent [14]. Multi-view fusion improves anatomical consistency and tumor core accuracy but requires three view-specific networks, increasing parameter count and inference latency relative to single-view axial processing [1], [1]. Feature-level or latent-space fusion (U-HVED, mmFormer) enables graceful degradation under missing modalities by embedding observed modalities into a shared representation, yet it demands careful alignment of inter-modal correlations and may still fail in extreme missing-modality scenarios [6], [20]. Attention-based fusion (ShaSpec, SMU-Net) adaptively recalibrates modality features, but its benefit is contingent on the quality of the modality-specific encoders and the availability of sufficient training data for the attention modules [15], [19]. Late fusion and ensembling deliver consistent but modest gains—approximately 1% Dice improvement with an ensemble of 10 models on BraTS 2018 [3]—at the cost of linear model replication and storage overhead; test-time augmentation yields only marginal improvements [3].

Missing-modality robustness paradigms present the most consequential deployment trade-offs. Dedicated models, trained separately for every modality subset, achieve high accuracy per subset but require 2^N − 1 models; for the four BraTS modalities this means 15 models, which is computationally prohibitive for training and storage [21], [15]. Catch-all models (HeMIS, U-HVED, ACN, mmFormer, ShaSpec, SMU-Net, M3AE) serve all subsets with a single network, reducing inference complexity [6], [15], [16], [18], [19], [20], [21], but may underperform dedicated networks in complete-modality settings unless reinforced with co-training or distillation [20], [21]. Modality imputation and synthesis (U-HVED, SMU-Net) enable joint completion and segmentation from a shared latent space, yet performance depends on synthesis fidelity and extreme missing-modality cases may yield unacceptable Dice scores [19], [20]. Co-training and knowledge distillation transfer knowledge from full-modal teachers to missing-modal students (ACN, SMU-Net, M3AE), but the student may still lag the teacher, and the design of adversarial or distillation objectives adds training complexity [16], [19], [21]. Masked autoencoding (M3AE) learns robust self-supervised representations under random modality and patch masking, but reported BraTS segmentation benchmarks for this paradigm remain limited [21].

Table 9. Strengths, limitations and unresolved challenges

Method family	Strengths	Limitations	Unresolved challenges	Representative studies [n]
CNN-based	Efficient local spatial modeling; low inference latency (25 s–3 min per brain) [12]	Bounded receptive field; limited long-range context [8], [12]	Scaling global context without prohibitive memory	[8], [12], [13], [17]
Transformer-based	Global anatomical context; strong Dice on BraTS 2020/2021 (e.g., Swin UNETR WT 0.927, TC 0.876, ET 0.853) [10]	High memory consumption; training complexity [8], [10]	Combining global modeling with missing-modality robustness	[8], [10], [22]
Hybrid CNN-Transformer	Balances global context and local boundary refinement [6], [10]	Increased architectural complexity; inconsistent gains across editions	Validating cross-edition generalization	[6], [8], [10]
Early (input-level) fusion	Simple implementation; standard for full-modality BraTS [14]	Degrades sharply with missing channels [14]	Graceful degradation under incomplete acquisitions	[14]
Multi-view fusion	Improves anatomical consistency and tumor core accuracy [1], [1]	Three view-specific networks; higher latency and storage [1], [1]	Reducing view redundancy without accuracy loss	[1], [1]
Feature-level / latent fusion	Enables graceful degradation under missing modalities [6], [20]	Requires careful inter-modal alignment [6], [20]	Aligning heterogeneous modalities in shared space	[6], [20]
Attention-based fusion	Adaptive modality recalibration [15], [19]	Depends on encoder quality and training data [15], [19]	Stabilizing attention under extreme missingness	[15], [19]
Late fusion / ensembling	Modest but consistent Dice gains (~1% with 10 models) [3]	Linear model replication; storage overhead [3]	Cost–benefit justification in clinical pipelines	[3], [12]
Dedicated models	High accuracy per modality subset [15], [21]	2^N − 1 models (15 for N = 4); impractical deployment [15], [21]	None—paradigm is inherently non-scalable	[15], [21]
Catch-all models	Single model for all subsets; reduced storage and inference complexity [6], [15], [16], [18], [19], [20], [21]	May underperform dedicated networks in complete-modality settings [20], [21]	Closing the accuracy gap with dedicated models	[6], [15], [16], [18], [19], [20], [21]
Imputation / synthesis	Joint completion and segmentation from shared latent space [19], [20]	Performance depends on synthesis fidelity; extreme cases may fail [19], [20]	Defining acceptable synthesis quality for diagnosis	[19], [20]
Co-training / distillation	Transfers full-modal knowledge to missing-modal students [16], [19], [21]	Student may lag teacher; complex adversarial/distillation design [16], [19], [21]	Matching teacher performance without full modalities	[16], [19], [21]
Masked autoencoding	Robust self-supervised representations under random masking [21]	Limited reported BraTS segmentation benchmarks [21]	Demonstrating segmentation gains on standard splits	[21]

6.2 Generalization, Robustness & Clinical/Deployment Concerns

Generalization across data sources remains uneven. EMMA reports similar DSC on BraTS 2017 validation and testing sets despite the latter containing data from different sources, indicating cross-site robustness [11]. By contrast, competing methods that fit the validation set well fail to retain performance on the test set, highlighting overfitting to site-specific

intensity distributions [11]. This is a direct barrier to multi-center clinical deployment, where scanner protocols and patient populations vary.

Automated segmentation accuracy remains inferior to expert inter-rater agreement [4]. Confounding factors routinely encountered in clinical imaging—blood products, post-operative air pockets or resection cavities, non-GBM entities, diffuse LGG boundaries, and cases lacking enhancing tumor sub-regions—are not systematically addressed by current architectures [4]. The BraTS evaluation convention further complicates interpretation: cases without enhancing tumor in the ground truth always yield Dice = 0 by definition, and postprocessing rules that exploit this convention can improve challenge ranking at the expense of introducing Dice = 0 failures in other cases [2], [13]. Such rank inflation undermines the clinical meaning of leaderboard positions.

Catch-all networks are economically attractive because a single model serves all modality subsets [21], yet the clinical consequences of modality-specific performance drops are uneven. Modality importance is asymmetric: FLAIR is the principal modality for WT, whereas T1c is critical for TC and ET [5], [14], [16], [18]. Missing FLAIR causes a sharp Dice decrease across all regions; missing T1c causes severe degradation for TC and ET; missing T1 or T2 causes only slight decreases [5], [16]. In extreme missing-modality scenarios, HeMIS and U-HVED achieve unacceptable DSC scores of approximately 10%–20% on ET and TC, whereas ACN maintains DSC above 40% [16]. These differences matter because ET and TC quantification are critical for assessing treatment response in high-grade glioma [5], [16]. Future work must establish minimum acceptable modality subsets for different clinical decision contexts (diagnosis, radiotherapy planning, follow-up), because the consequences of losing T1c are more severe than losing T1 or T2 [5], [16].

6.3 Limitations of this Review

This is a corpus-based narrative review rather than an exhaustive systematic review. Sources were searched in a curated internal PDF repository; the catalog contained 31 papers, of which 22 distinct papers were retrieved as evidence. No independent title/abstract screening or PRISMA flow was performed beyond retrieval relevance, so selection bias cannot be excluded. Furthermore, no published method was independently re-run; all quantitative claims are syntheses of reported evidence from the source documents, with the comparability constraints noted in Section 5. Consequently, relative rankings across BraTS editions should be treated as indicative rather than definitive, because units alternate between ratios and percentages, splits vary between training, validation, and testing, and editions differ in patient cohorts and annotation protocols [1], [2], [3], [7], [8], [10], [11], [12].

6.4 Open Research Directions

The following research directions are grounded in the evidence and follow from the research questions in Section 1:

1. **Robust segmentation without T1c (RQ 2).** Develop networks that maintain high Dice on TC and ET when T1c is missing, given its principled role in enhancing tissue visualization [5], [16]. Current catch-all models still exhibit severe degradation for TC and ET under T1c absence [5], [16].
2. **Computational budgeting for catch-all models (RQ 2).** Determine the optimal computational budget for single-model missing-modality systems in resource-constrained settings, considering the observed 25-second-to-3-minute inference range across CNN architectures [12] and the memory costs of Transformer backbones [8], [10].
3. **Transformer robustness without prohibitive memory (RQ 1 and RQ 2).** Investigate whether Transformer-based global modeling (mmFormer, Swin UNETR, TransBTS) can be combined with missing-modality robustness without prohibitive memory costs [6], [8], [10].
4. **Evaluation metrics for absent enhancing tumor (RQ 3).** Redesign metric aggregation to account for cases with absent enhancing tumor sub-regions and avoid rank inflation via postprocessing [2], [13].
5. **Standardized multi-metric reporting (RQ 3).** Standardize reporting of sensitivity, specificity, and Hausdorff distance alongside Dice to capture over- and under-segmentation tendencies [9].
6. **Minimum acceptable modality subsets (RQ 2 and RQ 3).** Define minimum acceptable modality subsets for different clinical decision contexts, because the clinical consequences of losing T1c differ from those of losing T1 or T2 [5], [16].

7. Conclusion

This survey synthesizes published evidence on deep learning fusion strategies and missing-modality robustness for multimodal MRI glioma segmentation in BraTS. Regarding RQ 1, CNN-based backbones provide efficient local modeling with low inference latency (25 seconds to 3 minutes per brain) [12], whereas Transformer-based methods capture global context and report strong Dice scores (e.g., Swin UNETR: WT 0.927, TC 0.876, ET 0.853 on BraTS 2021 testing) [10], and hybrid designs attempt to balance both properties [6], [10]. Regarding RQ 2, catch-all models offer the best deploy-

ment trade-off by serving all modality subsets with a single network [6], [15], [16], [18], [19], [20], [21], but dedicated models remain more accurate per subset and are impractical at scale (15 models for four modalities) [15], [21]; among catch-all techniques, ACN maintains DSC above 40% in hard missing-modality cases while HeMIS and U-HVED drop to approximately 10%–20% on ET and TC [16], and mmFormer reports an average 19.07% Dice improvement with only one available modality [6]. Regarding RQ 3, leading methods report strong within-edition performance (e.g., DKFZ CNN mean Dice WT 0.858, TC 0.775, ET 0.647 on BraTS 2017 testing [2]; TransBTS Dice WT 90.09, TC 81.73, ET 78.73 on BraTS 2020 validation [8]), but cross-edition comparison is constrained by differing units, splits, and cohorts [1], [2], [3], [7], [8], [10], [11], [12].

The outlook is cautious. Benchmark performance has advanced substantially, yet automated segmentation remains inferior to expert inter-rater agreement [4], and clinical deployability is limited by missing-modality fragility, uneven generalization across data sources [11], and evaluation artifacts linked to absent enhancing tumor cases [2], [13]. Closing the gap will require robustness-first architecture design, clinically meaningful minimum modality subsets, and standardized reporting that pairs Dice with sensitivity, specificity, and boundary metrics [9]. Future work should prioritize the open directions in Section 6.4 rather than pursuing incremental leaderboard gains.

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